ENGINEERING LEARNING ACADEMY

Engineering Masterclass 001

Career Intelligence Copilot — FR-016 Multi-Agent Orchestration

Functional Requirement	FR-016 — Multi-Agent Orchestration
Edition	Engineering Learning Academy — Lean Edition
Status	
Audience	
Document type	Official study edition (PDF)

Faithful rendering of the Lean Engineering Masterclass Markdown. Canonical engineering remains the FR acceptance report and ADR.

Contents

Executive Summary

FR-016 explored when multiple AI specialists are genuinely justified by authority boundaries rather than role names. The accepted design combined a Deterministic Orchestration Supervisor (DOS), the existing Bounded Opportunity Preparation Agent (BOPA), and a read-only Operational Briefing Specialist (OBS). The milestone concluded with **GO AS LEARNING PROOF ONLY**: direct BOPA remains the preferred daily workflow while orchestration provides a validated architectural foundation.

Engineering Story

FR-015 already solved everyday preparation. FR-016 therefore asked a different question: when does a second specialist create real engineering value? Persona-based splits (Preparation, Truth, Review agents) were rejected as multi-agent theatre because they added complexity without distinct permissions. The accepted solution separated coordination (DOS), bounded preparation (BOPA), and read-only operational synthesis (OBS).

Architecture Overview

- DOS: observe, delegate, audit, stop; no domain authority.
- BOPA: bounded preparation, verification and truth validation.
- OBS: read-only operational briefing.
- DelegationPolicy controls specialist selection.
- ToolPolicy controls specialist actions.

Key Engineering Principles

- Authority boundaries justify specialists.
- Supervisors coordinate without inheriting permissions.
- Typed handoffs are safer than conversational delegation.
- Deterministic orchestration is preferable when routing is knowable.
- Technical success does not automatically justify product adoption.

Trade-offs and Validation

Alternative multi-agent designs were rejected because they lacked meaningful authority separation. Validation demonstrated safe deterministic orchestration and permission isolation. The engineering outcome succeeded, but commercial value remained intentionally limited.

Interview Preparation

Q: Why introduce multiple agents? A: To separate authority, not merely responsibilities.

Q: Why is DOS deterministic? A: Routing was fully expressible from typed state, improving safety and repeatability.

Q: Why is OBS read-only? A: Independent analysis is stronger when it cannot modify the state being assessed.

Q: Why wasn’t orchestration made the default? A: Direct BOPA remained simpler for everyday preparation.

Three Things To Remember

1. Specialists are justified by authority boundaries.
2. Supervisors coordinate rather than execute.
3. Simpler architectures remain preferable until additional complexity earns its place.